SLoRa: A Systematic Framework for Synergic Interference Resilience In LPWAN

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***Abstract*—By offering long-range wireless connections with low energy consumption, Low-Power Wide Area Networks (LPWANs) have been widely deployed in many critical Internet-of-Things (IoT) applications. However, LPWAN's low-power nature leads to inevitable packet corruption under cross-technology interference (CTI) emitted from other wireless technologies that coexist in the same unlicensed band. To address this fundamental issue, researchers propose multiple lines of solutions, such as signal analysis and coding. Despite these efforts, the research on CTI mitigation needs to go one step further - to design a systematic framework for enhanced interference resilience.**

This paper introduces SLoRa, a systematic framework integrating symbol recovery with soft decoding for synergic interference resilience. Without loss of generality, we focus on LoRa, the dominating LPWAN technology. SLoRa's symbol recovery examines the corrupted signal through a two-stage analysis to recover incorrect LoRa symbols accurately at a low cost. To estimate the confidence of recovered symbols, SLoRa examines the statuses of LoRa and CTI, and it further leverages the estimated confidence via soft decoding to improve the error correction capability. We analyze SLoRa theoretically and evaluate it with real testbeds, including SX1280 LoRa chips and USRP B210. Experiments across multiple settings, e.g., different CTI sources and frequency bands, demonstrate that SLoRa effectively enhances interference resilience at a low cost. Compared with the state-of-the-art technique, SLoRa improves the CTI protection capability by 1.4x while only requiring 1% computation cost.

***Index Terms*—LPWAN, LoRa, Wireless Interference**

I. Introduction

Pervasive Internet of Things (IoTs) require ubiquitous connectivity with a long communication range. In contrast to wireless technologies, such as WiFi, ZigBee, and Bluetooth, that have limited transmission distances, the recently introduced Low-Power Wide Area Networks (LPWANs) offer long-range connectivity at low cost and low power. These unique features empower the rapid deployment of LPWANs in various applications, such as smart city [1] [2], environmental monitoring [3], [4], intelligent transportation [5] [6], and automated manufacturing [7]. Based on the operating frequency, the LPWAN spectrum could be classified as the licensed or unlicensed industrial, scientific, and medical (ISM) band. Specifically, NB-IoT and LTE-M require access to the licensed spectrum at extra cost. In contrast, other LPWAN technologies (e.g., LoRa) utilize the unlicensed ISM band to reduce cost and enable rapid deployment. According to Semtech, LoRa accounts for $> 50\%$ of the deployed LPWAN devices in 2026 [8].

However, the choice of the unlicensed spectrum leads to cross-technology interference (CTI). According to [9]–[11], LoRa's packet rate reaches 50% due to LoRa's inherent long-range communication distance, low energy consumption, and long packet duration. This severe packet loss leads to significantly increased delay and larger energy consumption.

Targeting this fundamental CTI problem, researchers have proposed multiple lines of solutions. Techniques like PSR [12] directly analyze the received signal under CTI to recover corrupted symbols. Although effective, it fails to recognize the reliability difference across recovered symbols - its recovery reliability decreases when a LoRa symbol is corrupted by longer CTI, while its heavy computation overhead is also unfriendly to low-power end devices. Forward error correction techniques, such as [13], [14], design customized coding schemes to protect LoRa from CTI. Since these methods have zero access to PHY-layer information, their reliability is directly limited by the selected coding method and coding rate. Until now, a systematic approach for enhanced interference resilience under CTI is still missing.

As a systematic framework, our SLoRa integrates symbol recovery with soft decoding. Its symbol recovery involves a two-step signal analysis in the amplitude and frequency domain, from the coarse to fine granularity. Its amplitude analysis first examines the statistical characteristics of LoRa and CTI to identify the signal segments that might contain CTI. Then its frequency analysis searches for LoRa's unique features in the frequency domain for CTI verification and symbol recovery. Via these lightweight procedures, SLoRa recovers corrupted symbols with better reliability but a much smaller cost than PSR [12]. For recovered symbols, SLoRa estimates their reliability confidence, considering relevant statuses of CTI and LoRa, such as the CTI duration with a LoRa symbol, signal-to-noise ratio (SNR), and spreading factors (SF). Specifically, SLoRa establishes a mathematical model for confidence estimation and validates this theoretical model with simulation. The estimated confidence is leveraged in SLoRa's soft decoding for interference resilience improvement. By doing so, SLoRa's systematic framework achieves synergic

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interference resilience under CTI at a low cost.

Specifically, SLoRa has the following contributions:

1) To the best of our knowledge, this is the first systematic framework to achieve LoRa's synergic interference resilience under CTI.
2) The design of SLoRa consists of lightweight symbol recovery and soft decoding. For symbol recovery, SLoRa proposes a two-step strategy of amplitude and frequency analysis. Specifically, SLoRa's amplitude analysis requires little computation, followed by its frequency analysis that examines LoRa's unique frequency-domain patterns with better reliability. After recovery of corrupted symbols, SLoRa estimates their confidence incorporating multiple factors and further exploits this estimation in soft decoding to enhance the protection capability. By doing so, SLoRa achieves synergic reliability for low-power LoRa communication against CTI at a low cost.
3) We evaluate our design with theoretical analysis and experiments via real testbeds. Extensive results demonstrate that SLoRa offers reliable protection against CTI. It achieves 1.4x interference resilience and is 100x faster than the state-of-the-art technique.

II. Motivation

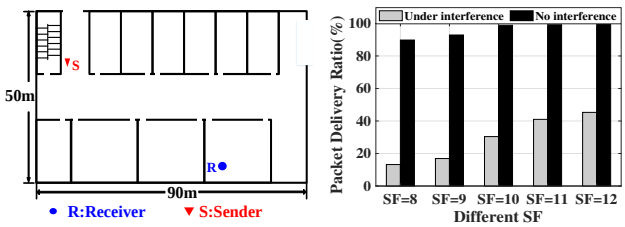
A. LoRa in the ISM Band

Targeting long-range communication for low-power devices, Semtech introduces the LoRa technique that adopts Chirp Spread Spectrum (CSS) for offering resilience against environmental noises. To reduce deployment costs, LoRa chooses to operate in the unlicensed industrial, scientific, and medical band (ISM band). Based on the ISM band frequency, LoRa works in sub-1GHz and 2.4GHz. However, the sub-1GHz ISM band has regional regulations. For example, the LoRa spectrum is 915 MHz for North America, 868 MHz for Europe, and 470 MHz for China. This discrepancy leads to extra hardware complexity for LoRa devices. To further enable rapid deployment, Semtech offers LoRa chips (SX1280 and SX1281) [15] in the 2.4GHz, a frequency widely available across the world. Additionally, the 2.4GHz LoRa shares the same modulation and coding procedures as the sub-1GHz LoRa, which enables a more extended transmission range compared with traditional IoT technologies (e.g., WiFi, Bluetooth, ZigBee).

B. CTI Impacts

Operation in the unlicensed ISM spectrum inevitably leads to CTI - low-power LoRa communication suffers from interference in other wireless technologies coexisting in the wireless band. For example, although the 2.4GHz band makes LoRa deployment more universal, it also introduces more serious CTI. That is because the 2.4GHz is crowded with popular technologies, like WiFi, ZigBee, and Bluetooth. And in the sub-1GHz, LoRa also could be interfered by transmissions of 802.11 ah. These CTIs generally have much larger energy and easily corrupt low-power LoRa communication.

To verify the impacts of CTI on LoRa transmissions, we conduct the experiments via real testbeds. Without loss of generality, we examine the LoRa transmission in the 2.4GHz and transmit 1000 packets by using the SX1280 device with different spreading factors (SF which decides the LoRa symbol duration and resilience). Fig. 1a shows the scenario map, we use SX1280 in different SFs from S to R in a long interior corridor, and transmit LoRa data packets with 12.5dBm transmission power. Due to the complex indoor environment and many obstructions, the SNR is close to -10dB. We transmit LoRa packets separately in the middle of the night with almost no WiFi interference and in environments with strong WiFi interference during the day. Fig. 1b demonstrates that the standard LoRa performs wells under the environment without interference - the successfully decoded 89.8% packets in SF8 and almost 100% packets when SF reaches 10. In contrast, the packet delivery ratio is only 13.2% under CTI when SF is 8. Even though the increase in SF enhances resilience, the packet reception rate is lower than 50%, much less than the case without interference. These experiments show the serious impacts of CTI on LoRa communication reliability.



(a) Indoor scenario (b) Packet delivery ratio

Fig. 1: (a) Indoor experiment scene; (b) PDR comparison.

III. System Overview

A. Overview

This work focuses on LoRa, which takes more than 50% of the LPWAN market share [15], for illustration purposes. Its design principles could be adopted for other unlicensed-spectrum LPWAN techniques. SLoRa consists of two major components: Lightweight Symbol Recovery and Soft Decoding. Fig. 2 demonstrates the system architecture.

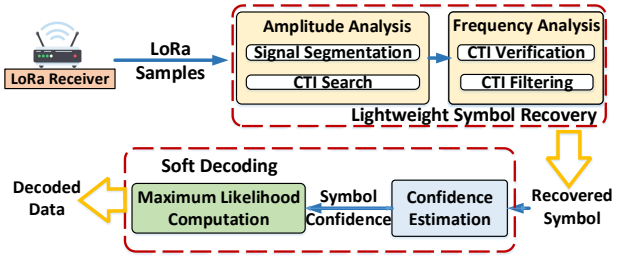


Fig. 2: Overview of SLoRa

Lightweight Symbol Recovery: SLoRa aims to directly analyze the corrupted LoRa signal from coarse to fine granularity in a cost-effective way for symbol recovery. Specifically, it first checks the amplitude changes with little computational overhead and then further adopts the frequency analysis that analyzes LoRa's unique signal characteristics. By searching

the signal in a hierarchical way, SLoRa manages to offer reliable symbol recovery at a low cost.

Soft Decoding: To fully boost the LoRa communication resilience under CTI, SLoRa also explores the soft-decision decoding with probabilistic demodulation outputs and a soft-input channel decoder for Hamming codes which is compatible with the existing standard LoRa encoding scheme.

B. Challenges

To offer a CTI-resilience framework with low cost and high reliability, we need to overcome the following challenges:

1. General capability. Since LoRa coexists with other wireless technologies in the ISM band, a LoRa symbol could be interfered by one or more wireless transmissions (e.g., WiFi, Bluetooth, ZigBee, and others). This requires us to design a unified protection framework applicable to different types of CTI without any prior knowledge, e.g., interference type, power, and duration.

2. Reliable protection. To alleviate CTI, existing research either analyzes the signal at the PHY layer or adds additional redundancy through coding at the upper layer. In contrast, our method needs to design a systematic framework for boosting the protection capability against CTI.

3. Low cost. Targeting low-power communication and applications, our method needs to offer reliable interference protection at a low computation and storage cost.

IV. Background

LoRa Modulation: LoRa uses the Chirp Spread Spectrum (CSS), which employs linear chirps to modulate data. According to the changing trend of the spectrum, it can be divided into upchirp and downchirp, whose frequency increases/decreases linearly with time. LoRa modulation is to use a series of upchirp signals with different initial frequencies to represent LoRa symbols, and each symbol carries the bit information of the transmitted data, as shown in Fig. 3a. The signal transmitted by LoRa can be expressed as:

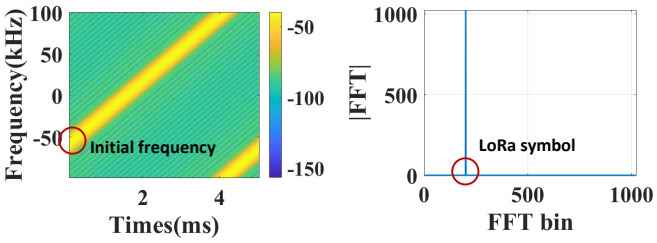
$$x_s[n] = A_s e^{j2\pi\left(-\frac{BW}{2}+sf_\Delta+nf_\Delta\right)\frac{n}{N}} \tag{1}$$

where A_s is the signal amplitude of the transmitted signal s, and the transmission bandwidth is generally from $-\frac{BW}{2}$ to $\frac{BW}{2}$. N is the total number of sampling points of one symbol and n represents the index of the discrete sequence. f_Δ is the frequency change rate which is equal to $\frac{BW}{N}$. The starting frequency of the symbol is $sf_\Delta - \frac{BW}{2}$, and the frequency gradually increases in steps of f_Δ.

LoRa Demodulation: The demodulation stage performs a dot multiplication of the received signal $x_s[n]$ with standard downchirp, which can be expressed as $x_0^*[n] = \mathrm{A_d} e^{-j2\pi\left(-\frac{BW}{2}+nf_\Delta\right)\frac{n}{N}}$ and $A_d = 1$. Then the signal becomes

$$x_s[n] \cdot x_0^*[n] = A_s e^{j2\pi(sf_\Delta)\frac{n}{N}} \tag{2}$$

In Eq.2, $x_s[n] \cdot x_0^*[n]$ contains only the signal frequency sf_Δ and then after the FFT, we can get the symbol s which is the frequency bin corresponding to the maximum value in the spectrum, as shown in Fig. 3b.



(a) LoRa signal (b) FFT spectrum

Fig. 3: LoRa demodulation: (a) is the initial LoRa symbol; (b) is the FFT spectrum after dechirping.

LoRa Coding/Decoding: The standard LoRa adopts forward error correction (FEC) with Hamming coding that supports 4 coding rates $\mathrm{CR} \in \{4/5, 4/6, 4/7, 4/8\}$. m is the length of the codeword after encoding $m \in \{5, 6, 7, 8\}$. When m is 5 or 6, it can only detect errors without correction capability. When m is 7 and 8, a one-bit error within a codeword could be corrected. In addition, LoRa adopts a diagonal interleaver to interleave the bits within each block, where the LoRa block length is $SF \times m$ bits, instead of the packet length. At the receiver side, it adopts corresponding de-interleaving and decoding procedures to combat interference.

V. Lightweight Symbol Recovery

This section introduces our lightweight symbol recovery technique. SLoRa first adopts amplitude analysis with little computation cost to locate suspicious signal parts that might be affected by CTI. Then it further analyzes them in the frequency domain for verification and recovers corrupted LoRa symbols based on the clean LoRa signal.

A. Amplitude Analysis

This section introduces SLoRa's low-cost amplitude analysis for identifying potential CTI segments within a LoRa symbol. Specifically, it first analyzes the amplitude changes of the received LoRa signal to cut it into multiple segments and further identify suspicious segments that might contain CTI. For illustration purposes, we focus on the scenario where CTI's energy is larger than LoRa, which is dominating given the LoRa's long distance and low power. When LoRa's energy is higher, LoRa's CSS reliably protects against CTI.

*1) **Signal Segmentation**:* For a LoRa symbol s that is corrupted by CTI, we denote its signal received at a LoRa receiver as $y_s[n]$.

$$\mathrm{y_s[n]} = \begin{cases} hx_s[n] + w[n] & n \in \mathcal{N}_1 \\ hx_s[n] + w[n] + I[n] & \mathrm{n} \in \mathcal{N}_2 \end{cases} \tag{3}$$

In Eq.3, $\mathcal{N}_1$ and $\mathcal{N}_2$ represent the set of clean chips and CTI-contaminated chips, respectively. n denotes the index within this LoRa symbol and $n \in \{0, 1 \ldots 2^{SF} - 1\}$. $w[n]$ represents the environment Gaussian white noise. Specifically, $w[n] \sim \mathcal{CN}\left(0, \sigma^2\right)$ $\sigma^2 = \frac{N_0}{2}$ and N_0 is the signal-sided noise power spectral density. $I[n]$ is the CTIs that could result from multiple wireless technologies/transmissions. Note that those $I[n]$ that can negatively affect LoRa demodulation will

basically have higher energy than LoRa. h represents the channel impulse response and could be regarded as a constant during a short time. Note that Section V-B further addresses the changes of h in the dynamic environment.

According to the standard LoRa demodulation, after dechirping by Eq.2, the clean LoRa signal becomes $\widehat{y_s[n]} = hA_s e^{j2\pi(sf_\Delta)\frac{n}{N}} + \widehat{w[n]}$. Specifically, $\widehat{w[n]}$ still accord with complex Gaussian white noise distribution, and its amplitude envelope $|\widehat{w[n]}|$ can be expressed as a Rayleigh distributed random variable. In addition, the clean LoRa signal amplitude spectrum $A = |\widehat{y_s[n]}|$ can be regarded as the sum of the original signal and the multipath noise obeying the Rayleigh distribution, which can be approximated with Rician distribution $f\left(A \mid h\bar{A}_s, \sigma\right)$. $\bar{A}_s$ is the mean amplitude of the signal and σ is the standard deviation of the noise. In contrast, the CTI has different communication protocols and thus has different amplitude patterns. This motivates us to distinguish clean LoRa signal from CTI, without prior knowledge of CTI.

Based on the above theoretical analysis, it is feasible to utilize the amplitude pattern changes to detect CTI. Specifically, we calculate the amplitude spectrum of the signal after dechirping and $x_0^*[n]$ is the standard downchirp,

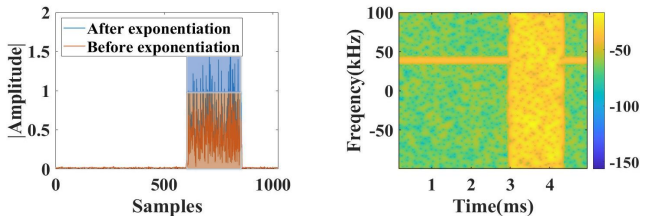
$$A_s[n] = abs\left(y_s[n] \cdot x_0^*[n]\right) \tag{4}$$

To avoid environmental noises and improve reliability, we adopt signal preprocessing and a window sliding algorithm to detect the amplitude change point.

• **Signal preprocessing.** Firstly, we normalize and exponentize the amplitude spectrum before starting the analysis.

$$A'_S[\mathrm{n}] = e^{\frac{\mathrm{A_S}[n]-\min(\mathrm{A_S}[n])}{\max(\mathrm{A_S}[n])-\min(\mathrm{A_S}[n])}} \tag{5}$$

The exponential function increases monotonically and the slope gradually increases, which can increase the contrast ratio of the amplitude value. As shown in Fig.4a, it can be seen that after the signal preprocessing of A_s, the CTI part is nearly doubled, and the part of the clean chips becomes smaller, which shows that it can increase the gap between clean chips and CTI so that the model can better identify interference.



(a) Signal preprocessing (b) Spectrogram of Figure (a)

Fig. 4: Comparison between clean LoRa signal and CTI

• **Window sliding.** The main idea of window sliding is to calculate the discrepancy index to measure the difference between the intercepted signal in the first half of the window win_1 and the second half of the window win_2. The discrepancy index is denoted as follows

$$d\left(As'_{a:t}, As'_{t:b}\right) = c\left(As'_{a:b}\right) - \left[c\left(As'_{a:t}\right) + c\left(As'_{t:b}\right)\right] \tag{6}$$

In Eq.6, $win = As'_{a:t}$ and $t = a + \frac{b-a}{2}$. $c(\cdot)$ is the cost function of measuring the number of samples that deviate from the empirical observation in this window. According to our experiments, the window size is set as $\frac{2^{SF}}{8}$ empirically.

Because the amplitude distributions of clean LoRa chips and the CTI are different, statistical features can be used to distinguish those signal parts. We utilize the Mean shift and Variance shift to calculate the cost function simultaneously.

$$c_M\left(A's_{a:b}\right) = \sum_{t=a+1}^{b} \left|A's_t - \bar{y}_{a:b}\right|^2 \tag{7}$$

$$c_V\left(A's_{a:b}\right) = (b-a)\log\sigma^2_{a:b} \tag{8}$$

where $\bar{y}_{a:b}$ and $\sigma^2_{a:b}$ are the empirical mean and variance of the segment $A's_{a:b}$. Using the mean and variance at the same time can prevent the model from being unable to divide the signal when a certain type of statistical feature is not obvious.

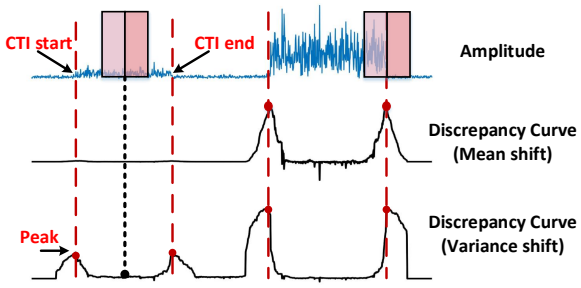


Fig. 5: Window sliding algorithm

Our previous analysis shows that the amplitude follows the Rice distribution when there is no CTI in this window. In contrast, CTI leads to a surge in the discrepancy index (Eq.6). The maximum points are the start and the end of CTI. This is because at the start and the end of CTI, the win_1 and win_2 cover signals with dramatically different distributions, leading to the highest discrepancy. Here we only emphasize the difference in distribution and do not pay more attention to the specific distribution of CTI, due to the lack of prior knowledge of CTI. Fig 5 illustrates one example of the received LoRa signal. It is clear that the discrepancy indexes reach the highest at the CTI boundary, which could be utilized to detect suspicious signal segments that might contain CTI. It can also be seen that using Mean shift alone may ignore low-energy CTI, but using variance and mean at the same time can make signal segmentation more robust.

Algorithm 1 shows the computation procedures for signal segmentation. Given the sliding window, whose midpoint is t, we first compute the discrepancy index $Z[t]$. Then we further find the change points (local maximum points) via peak searching. Fig. 6 demonstrates the example of signal segmentation. Based on the discrepancy computation results, the whole signal is divided into 5 segments for further analysis.

*2) **CTI Search**:* After dividing the received LoRa signal into multiple segments, we further search for CTI via the following procedures. The basic idea is to examine the statistical

Algorithm 1: Window sliding

Input: Amplitude spectrum A_s, Cost function $c(.)$, Half-window width ω, Peak search procedure $findpeaks(.)$.

Initialize Z $\leftarrow$ zeros(1,2^{SF});
for $t = \omega, 2\omega \ldots 2^{SF} - \omega$ **do**
  $p \leftarrow t - \omega : t$;
  $q \leftarrow t : t + \omega$;
  $r \leftarrow t - \omega : t + \omega$;
  Z[t] $\leftarrow$ c $(A_p(\mathrm{r}))$ − c $(A_p(\mathrm{p}))$ − c $(A_p(\mathrm{q}))$;
end
$L \leftarrow findpeaks(Z)$;

Output: L, the change point indexes.

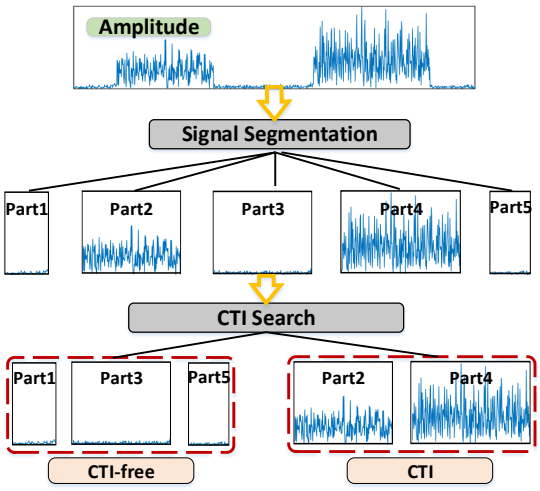


Fig. 6: SLoRa amplitude analysis example

values in each segment and then classify these segments into the CTI category and CTI-free category.

Specifically, we first calculate the mean value of each segment. Based on these mean values, we add the segment with minimum mean value to the CTI-free set, while adding the segment with maximum mean value to the CTI set. Then we use the nearest neighbor idea and iteratively pick one segment and compute the distance to the CTI set and CTI-free set, respectively. Note that our goal is to ensure that the segments in the CTI-free set are LoRa signals, and could tolerate a LoRa segment to be classified into the CTI set, since Section V-B analyzes LoRa frequency features with high reliability. As a result, we add an additional penalty α when computing the distance to the CTI-free set, and this penalty is calculated from the min and maximum mean values.

In summary, the window sliding algorithm emphasizes the difference between the amplitude distribution of pure LoRa and CTIs, and there is no requirement for any prior knowledge about CTI, e.g., interference type, power, and duration. At the same time, SLoRa uses a variety of statistical features (mean shift and variance shift) to detect the CTI, making the results of the model more accurate and robust, which can be used in various complex situations.

B. Frequency Analysis

With the signal segments identified by amplitude analysis, our frequency analysis examines LoRa features for further verification, which offers better reliability with a larger cost.

*1) **CTI Verification**:* The amplitude analysis cuts the received wireless signal into multiple segments and searches the suspicious segments corrupted by CTIs. The number of segments is constantly changing depending on the number of CTI parts that the symbol is disturbed by. However, due to the long symbol duration and the potential dynamic environment, the LoRa signal energy might change dramatically in practice, which might lead to inaccurate classification. Fig. 7 shows the received LoRa signal within a LoRa symbol without CTI. It can be seen that the signal energy drops significantly due to the environment change. Since SLoRa examines the amplitude changes in Section V-A, it might inaccurately classify a CTI-free segment as a CTI segment.

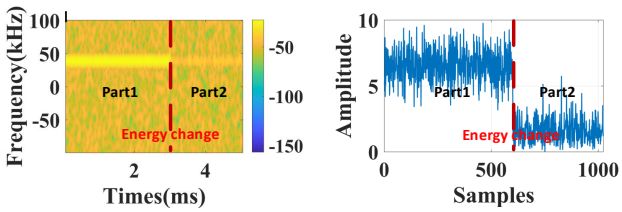


Fig. 7: Symbol energy change example (without CTI)

To offer more reliable CTI detection, we further examine the LoRa signal features in the frequency domain, which is completely different from other types of interference. Specifically, we define the Single Frequency Factor Coefficient (SFFC).

$$SFFC = \frac{R^2_{\max}[k]}{\sum_{k=1}^{K} R[k]^2} \tag{9}$$

where $R[k] = FFT(y_s[k] \cdot x_0^*[k])$ and represents the FFT results of the current signal segment generated in Section V-A.

Since a standard LoRa symbol adopts CSS modulation technique, it has a dominant frequency component in the frequency spectrum - this frequency component has a much larger value than all other components and is recognized by the LoRa demodulation result. In this case, the SFFC result is high. For a segment corrupted by CTI, the frequency components after FFT are diverse, lacking a dominating frequency, since other wireless communication techniques are not based on CSS. Under this circumstance, SFFC changes significantly.

Motivated by this, we calculate the SFFC for each segment marked as CTI in Section V-A. For a segment marked as CTI and has a high SFFC, it is re-classified as CTI-free. Note that the SFFC result also depends on the segment length and also SNR. For a CTI-free segment, the SFFC result increases with the increase of segment length and SNR.

Since we first rely on amplitude analysis in Section V-A to select segments that are possibly affected by CTI, our SFFC examination is only applied to those segments in the CTI set to reduce the probability of the CTI-free part being misclassified. In practice, it is possible that some signal segments are affected

by CTI but are classified as CTI-free sets. This will not affect SLoRa, since this CTI's energy is generally very small and could be mitigated by standard LoRa demodulation. The two-stage analysis offers good reliability across different scenarios as validated in Section VIII.

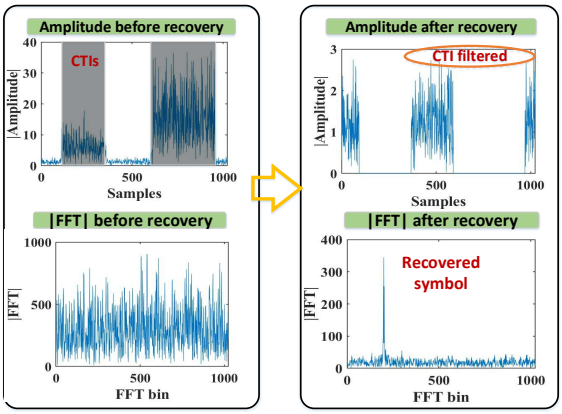


Fig. 8: SLoRa recovery result

Fig. 8 shows an example of a corrupted LoRa signal. Our SLoRa first segments this LoRa signal via amplitude analysis. Then it applies frequency analysis to verify the CTI segments, followed by the symbol recovery in the next section.

*2) **CTI Filtering**:* After identifying LoRa chips corrupted by CTI, we further filter out the CTI for symbol recovery.

$$\mathrm{c_s[n]} = \begin{cases} hx_s[n] + w[n] & n \in \widehat{\mathcal{N}_1} \\ 0 & \mathrm{n} \in \widehat{\mathcal{N}_2} \end{cases} \tag{10}$$

where $\widehat{\mathcal{N}_1}$ and $\widehat{\mathcal{N}_2}$ are the sets of estimated CTI-free and CTI chips, respectively. Similar to the traditional demodulation method, we set the CTI part to 0. Then we follow the standard LoRa demodulation procedures for the symbol recovery.

Fig. 8 shows the corrupted LoRa signal and its demodulation result. It can be seen that the FFT result of the original signal lacks a dominating frequency component, meaning the demodulation fails. When SLoRa filters out CTI, the FFT result demonstrates a high peak, which is exactly the transmitted LoRa symbol. SLoRa successfully recovers the corrupted LoRa symbol via its amplitude and frequency analysis.

VI. Soft Decoding

Although SLoRa analyzes the corrupted signal for symbol recovery, its performance is inevitably affected by the CTI ratio within a LoRa symbol. SLoRa will fail to demodulate a symbol that is affected by CTI with a long duration. (more details in Section VII and Section VIII-A). As a result, SLoRa adopts the soft decoding for further communication resilience under CTI. It first examines LoRa and CTI characteristics for confidence estimation of recovered symbols, followed by the maximum likelihood computation. Since the standard LoRa packets already adopt hamming coding [16], SLoRa directly utilizes this opportunity in its soft decoding without designing its own decoding schemes at the application layer. This ensures that SLoRa is directly compatible with the existing LoRa packet format and applications. Note that SLoRa's general design philosophy could be directly applied to other customized coding solutions directly by combining the estimated symbol confidence with the corresponding decoding procedures.

*1) **Confidence Estimation**:* According to the CSS demodulation, the frequency bin with the highest amplitude will be selected as the symbol's value after performing dechirping and FFT, which is due to the orthogonality of the LoRa signal [17]. Specifically, with the received signal $y_k[n]$ that corresponds to a transmitted waveform $x_k[n]$ over an **AWGN + CTI** channel, the correlator outputs of the LoRa demodulator is

$$\sum_{n=0}^{2^{SF}-1} \mathrm{y_k}[n] \cdot x_i^*[n] = \begin{cases} 2^{SF}\left(h\mathrm{A_i} + \varphi_i + CTI_i\right) & i = k \\ 2^{SF}\left(\varphi_i + CTI_i\right) & i \neq k \end{cases} \tag{11}$$

where h and A are the channel gain and amplitude during the transmission of symbol k. φ_i is the complex Gaussian noise process and CTI_i stands for the cross-technology interference.

Intuitively, the demodulation correctness is related to channel quality, especially the noise power and CTI degree. The higher the energy of a compensated frequency peak, the higher the chance that the frequency peak carries the transmitted symbol. So we can further define $\delta_i = |\varphi_i + CTI_i|$, which is the magnitude of the envelope of noise and CTI. Consequently, the probability of a LoRa symbol error can be expressed as

$$P_{e|k} = \Pr\left[\max_{i,i\neq k}\left(\delta_i\right) > |h\mathrm{A_k} + \varphi_k + CTI_k|\right] \tag{12}$$

CTI power is much higher than LoRa power, which greatly increases the error probability. However, by the lightweight symbol recovery, SLoRa can accurately identify CTI, filter it out and further can calculate the CTI_{ratio} for each LoRa symbol. Therefore, the clean chips obtained through SLoRa can be approximated to the signal transmitted in **AWGN** channel, and the probability of symbol error becomes

$$P'_{e|k} = \Pr\left[\max_{i,i\neq k}\left(\delta'_i\right) > |h\mathrm{A_k} + \varphi_k|\right] \tag{13}$$

Furthermore, $\delta'_i = |\varphi_i|$ is a random variable that satisfies the Rayleigh distribution, and its cumulative distribution function (CDF) is $F_{\delta'_i}\left(\delta'\right) = 1 - \exp\left[-\frac{\delta'^2}{2\sigma^2}\right]$. Let $\rho = \max_{i,i\neq k}\left(\delta'_i\right)$ depict a random variable for the maximum of $2^{SF}-1$ i.i.d. Rayleigh random variables [18]. The cumulative distribution function for ρ can be given as

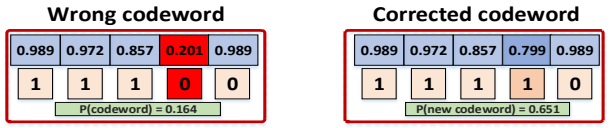
$$F_\rho(x) = \left[1 - \exp\left[-\frac{x^2}{2\sigma^2}\right]\right]^{2^{SF}-1} \tag{14}$$

Given Eq.13 and 14, the amplitude $|h\mathrm{A_k} + \varphi_k|$ of a clean LoRa chips obeys the Rice distribution $f\left(A \mid h\bar{A_k}, \sigma\right)$. $\sigma^2 = \frac{N_0}{2}$ and N_0 is the signal-sided noise power spectral density. Our design assumes that each LoRa symbol has the same probability, the average symbol confidence P_c can be expressed as

$$\mathrm{P_c} = 1 - P'_{e|k} = 1 - \int_0^\infty \left[1 - F_\rho(x)\right] f_{A_k}(x) dx \tag{15}$$

The confidence P_c is jointly decided by noise power N_0 and symbol energy (which can be combined into SNR), as well as CTI_{ratio} that is calculated in SLoRa's symbol recovery stage.

Here, considering the erroneous symbols must contain erroneous bits and the SNR corresponding to the pure chips keep relatively stable within a short duration, we directly set the confidence corresponding to SF bits in one symbol to be the same. With the estimated confidence for each LoRa symbol and their bits, SLoRa further performs de-interleaving to convert them to the bit confidence within a codeword. Due to illustration purposes, these details are omitted. Fig. 9 (a) shows an example of the bit confidence within a wrong codeword with the coding rate $4/5$. The red bit is wrong due to CTI and its confidence is lower than other bits.



(a) Wrong codeword (11100) with its bit confidence (b) Corrected codeword (11110) with its bit confidence

Fig. 9: Soft decoding example

*2) **Soft Decoding**:* SLoRa adopts soft decoding [19] via the maximum likelihood searching for additional error correction capability. Denote the codeword as R and the ith bit within R as b_i. SLoRa searches the codeword space C to compute the decoding result. Specifically, it checks all the possible patterns to select a codeword C_j with the highest probability.

$$P\left(R \mid C_j\right) = \text{argmax}_j \prod_{j=1}^{n} P\left(b_i = C_{ij} \mid C_{ij}\right) \tag{16}$$

To reduce the search time, SLoRa further adopts a heuristic search. When a LoRa symbol suffers from more CTI, its bits are likely to be incorrect, which could be detected through the low confidence estimated. Examples of confidence distribution for incorrect and correct demodulation results are demonstrated in Fig. 14. As a result, SLoRa starts with bits with low confidence. It iteratively flips these bits to generate codewords in the codeword space and outputs the codeword with the best probability as the decoding result. Fig. 9 (b) shows the result of soft decoding for Fig. 9 (a). SLoRa selects a codeword with the highest confidence depicted and successfully corrects the corrupted symbol. SLoRa's error correction is related to the coding rate (CR). Traditional Hamming decoding cannot correct errors when CR=4/5 and 4/6. SLoRa can correct one-bit errors when CR=4/5, and two-bit errors when CR=4/6. More bit errors might lead to SLoRa's decoding errors.

VII. Theoretical Analysis

A. Accuracy Analysis

To evaluate the symbol recovery accuracy and confidence estimation (Eq.15), we utilize simulation and send 10,000 LoRa symbols randomly under different CTI ratios and the same SNR (5dB). Fig. 10 shows symbol recover confidence

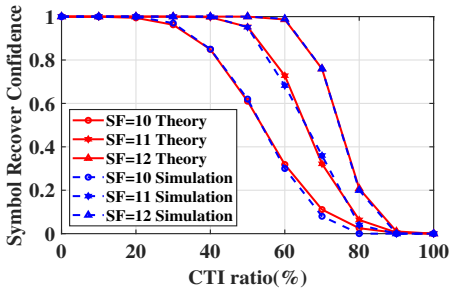


Fig. 10: Symbol recover confidence with different CTI degrees

values calculated by SLoRa are closely consistent with the successful symbol demodulation ratios offered by simulation. Furthermore, with an increasing proportion of CTI, the symbol recovery resilience decreases. In addition, the increase of SF makes LoRa symbol a higher chance of successful recovery, because the larger SF, the longer the duration, the stronger the anti-interference ability, so as to provide better resilience.

B. Cost Analysis

Time complexity. At the PHY layer, SLoRa first adopts amplitude analysis, which has the complexity of approximately $O(N)$. Then in frequency analysis, the complexity is $O(N \times logN)$. Moreover, SLoRa adopts soft decoding. Due to LoRa coding rules, there are at most 16 codeword combinations in the decoder, so the decoding complexity is $O(N)$. In summary, the overall complexity is $O(N+NlogN)$.

Space complexity. SLoRa requires storing the amplitude change of and FFT results. A symbol has N chips. Since the amplitude spectrum and FFT operations do not perform simultaneously, we choose the maximum storage space that satisfies both operations, which is a $1 \times N$ double array.

In contrast, the time complexity of PSR [12] scheme is $O\left(N^2 \times \log N\right)$, and it requires the storage space of approximately N^2 double matrices. For the standard LoRa, the demodulation cost is only $O\left(N \times \log N\right)$. And it needs a $1 \times N$ double array to store the FFT result. From what has been discussed above, SLoRa has a low time complexity and low storage space, which are friendly to low-cost LoRa devices.

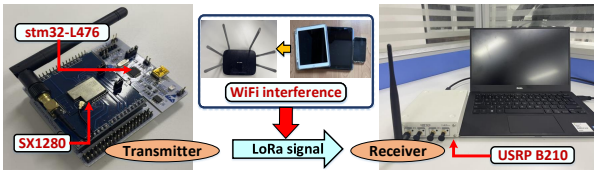
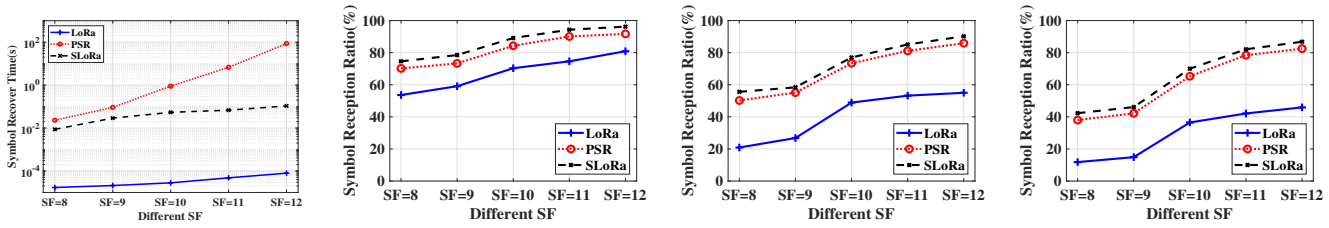


Fig. 11: Evaluation setup

VIII. Evaluation

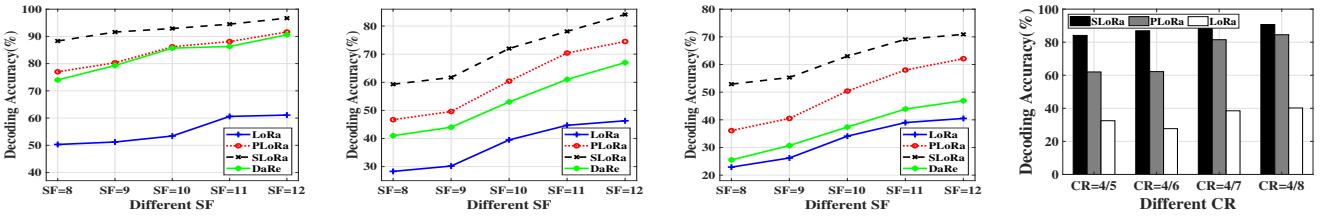
This section presents the evaluation of SLoRa via real testbeds shown in Fig. 11. Our transmitter includes an SX1280 LoRa transceiver chip and stm32-L476 controller, for transmitting LoRa packets. The receiver utilizes USRP B210 to capture the wireless signal at the normal LoRa sampling rate.

To evaluate the resilience of SLoRa, we conduct extensive experiments with different settings under the typical wireless interference at 2.4GHz which is the frequency band



(a) Symbol Recover Time (b) SRR (Low Interference) (c) SRR (Medium Interference) (d) SRR (High Interference)

Fig. 12: Demodulation performance at the PHY layer compared with the standard LoRa [15] and PSR [12]. (a) Computation time; (b)-(d) Symbol reception ratio under low, medium and high interference



(a) Decoding Accuracy (Low Interference) (b) Decoding Accuracy (Medium Interference) (c) Decoding Accuracy (High Interference) (d) Decoding Accuracy Under Different CRs

Fig. 13: Decoding performance compared with standard LoRa [15], PSR [12] and DaRe [13]. (a)-(c) Decoding accuracy under low, medium and high interference; (d) Decoding accuracy under different coding rates

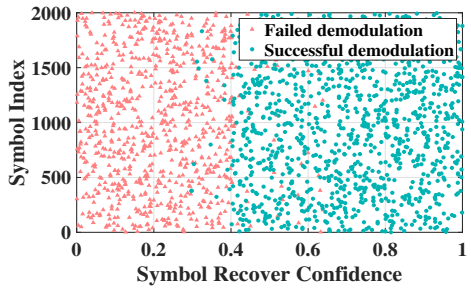


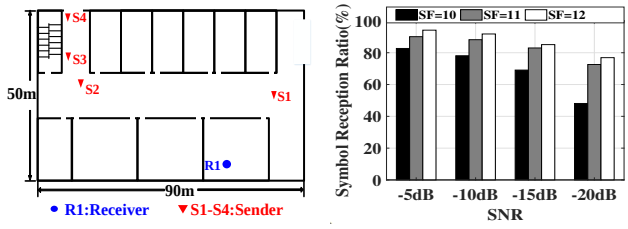
Fig. 14: Symbol recovery confidence vs demodulation results of SX1280. We emphasize indoor experiments since indoor scenarios (especially the 2.4GHz) generally have more severe CTI. As the WiFi is the dominating interference in the 2.4GHz, our evaluation focuses on the CTI due to WiFi traffic, while this section also analyzes the SLoRa's performance under different CTI types like ZigBee and Bluetooth and other LoRa operating frequency bands like 915MHz and 433MHz.

A. Demodulation Accuracy

This section analyzes the protection of symbol recovery. Specifically, we first compare SLoRa with standard LoRa and PSR, the state-of-the-art CTI resilience technique [12], and then examine SLoRa under different parameters.

a) Compared with existing works. For controlled experiments, we utilize commercial WiFi cards (phones, iPads, computers, etc.) to generate different volumes of WiFi interference. The numbers of operating WiFi devices are 1, 3, and 5 for low, medium, and high interference. More than 2,000 LoRa packets (100 Bytes) are transmitted. The transmit power is 12.5dBm. Considering that in LoRa's low-power and long-range features, we choose SNR = -5dB. Note that the impacts of SNR are extensively studied in the later part of the evaluation.

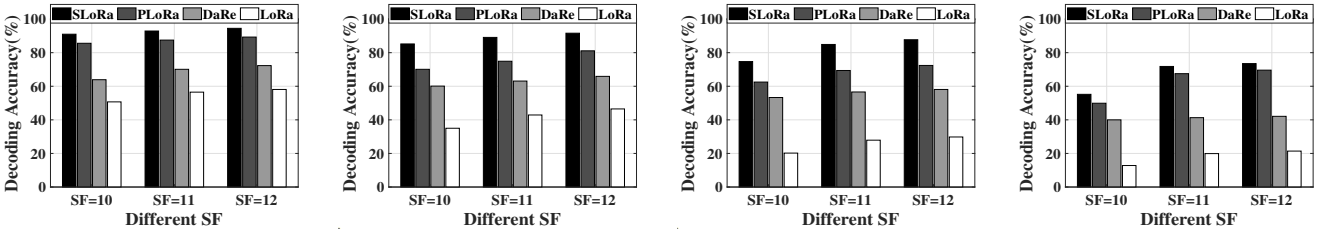
Fig. 12 shows the Symbol Reception Ratio (SRR) under different SF parameters and volumes of WiFi interference and the Symbol Recover Time (SRT) of the above three methods. It is clear that SLoRa enables reliable symbol recovery. For example, under the high interference, SLoRa's SRR is 69.9% when the SF value is 10, compared with 36.5% in LoRa. A large number of experiments show that SLoRa provides reliable symbol recovery under different CTI scenarios and SF parameter settings. On the other hand, our design significantly improves the computation speed while ensuring symbol recovery performance compared with PSR. From Fig. 12a, SLoRa demonstrates superior computation advantage. When SF=10 in high interference, the SRR of SLoRa is basically the same as that of PSR. However, our symbol recovery time is 16.56x faster than PSR. As SF increases, the processing time of PSR shows a cliff rise while SLoRa only requires **less than 1%** computation cost. When SF=12, PSR needs 86.28s to restore a symbol, while our design only needs 0.11s.



(a) Indoor experiment scene (b) SRR under different SNR

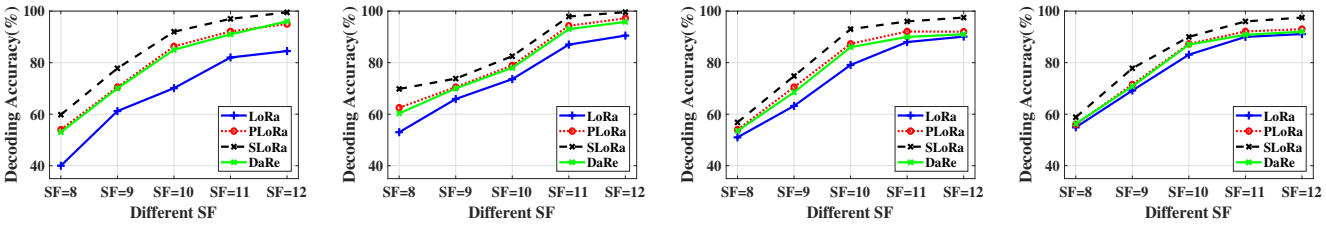
Fig. 15: (a) Indoor experiment scene. LoRa senders are placed from near to far at points S1-S4 respectively under different SNR; (b) Symbol recover performance under different SNR.

b) Performance under different parameters. SNR and SF are two main parameters that influence LoRa demodulation



(a) Decoding Accuracy (-5dB) (b) Decoding Accuracy (-10dB) (c) Decoding Accuracy (-15dB) (d) Decoding Accuracy (-20dB)

Fig. 16: Decoding accuracy under different SNR



(a) Zigbee (b) Bluetooth (c) 915MHz (d) 433MHz

Fig. 17: Decoding accuracy under different CTIs and other LoRa working frequency bands, such as 915MHz and 433Mz, which could be corrupted by 802.11.ah, telemetry transmitters, other LPWAN technologies such as Weightless and ZETA.

accuracy. So we also evaluate the robustness of SLoRa under different SNR and SF. The channel quality is changed by changing the distance between the transmitter and receiver, and their corresponding SNRs are -5dB, -10dB, -15dB, and -20dB. The further the transmitter is from the receiver, the lower the SNR. Fig. 15b shows the relationship between the symbol reception ratio and SNR. Under the same SNR, the symbol reception ratios of SF11 and SF12 are higher than that of SF10. In the case of good channel quality, SLoRa performs well, basically receiving more than 90% of the symbols under SF11 and 12. With the decrease in channel quality, the symbol reception ratio also decreases even with the deployment of our system. This is mainly because when LoRa symbols are submerged by noise, the clean chips might not be enough for recovery. This situation is particularly obvious when SF is small. When SF=10 and SNR is -20dB, only 48.1% symbol can be received, and the corresponding symbol reception ratios of SF=11 and 12 are 72.5% and 80.1%, respectively.

B. Decoding Accuracy

This section demonstrates the additional protection of soft decoding on top of symbol recovery. We compare the decoding accuracy of SLoRa with the standard LoRa, PLoRa (PSR's demodulation method combined with the traditional LoRa decoding), and DaRe [13] (an application-layer coding scheme).

a) Estimated confidence for recovered symbols. We evaluate the estimated symbol confidence under CTI. Without loss of generality, we randomly select 2000 symbols (SF=10) under different levels of CTI. Fig. 14 shows the demodulation results (if the transmitted symbol could be correctly demodulated) with their specific estimated confidence. It is clear that symbols with higher confidence are more likely to be demodulated correctly, which will be leveraged in soft decoding.

b) Performance with different interference levels. Fig. 13 illustrates the decoding performance of different interference levels and SF parameter settings with a CR of $4/5$. Without loss of generality, the SNR is -5 dB and the LoRa packet length is 100 bytes. The comparison of SLoRa, PLoRa, DaRe, and LoRa shows that SLoRa is more suitable for CTI protection. In the low interference state, the LoRa packet is still affected by CTI. For instance, when the SF value is 10, the decoding accuracy of LoRa is only 52.8%. Via PLoRa, we successfully improve the decoding accuracy to 86.5%. DaRe basically provides the same packet recovery performance as PLoRa in the low interference state. The performance gain comes from the improved symbol reception ratio. On this basis, the SLoRa with a soft decoding algorithm can further improve the decoding performance to 92.9%. With the increase of interference, LoRa suffers from more packet corruption, and the ability of DaRe and PLoRa to recover packets gradually decreases. But our design still maintains relatively stable data recovery performance. As shown in Fig. 13c, under high Wi-Fi interference and SF10, LoRa can only successfully decode 37.1% of the data. DaRe only improves the accuracy of demodulation by about 4% and more than 50% of the data can be decoded through the PLoRa. SLoRa can further improve the decoding accuracy to 63%.

c) Performance with different SNRs. To examine our system more thoroughly, we also conduct experiments to measure the decoding accuracy of SLoRa, PLoRa, DaRe, and LoRa with different SNR and SF under the same CTI level. From Fig. 16, it is clear that larger SF leads to higher decoding accuracy on all SNR levels. This is because larger SF means longer duration, thus offering better resilience. We also observe that, although extremely low SNR seems a great challenge to LoRa decoding, SLoRa can achieve high decoding accuracy with large SF. For example, when the SNR is -20dB, the decoding

accuracy for SF11 in LoRa is only 19%, while more than 70% bytes can be decoded successfully at SF$\geq$11 in SLoRa.

d) Performance of different CR. The factors that affect the decoding performance of LoRa are not only the experiment scene changes but also the parameter setting, especially the coding rate. So we explore the impact of different coding rates on SLoRa by setting coding rates as 4/5, 4/6, 4/7, and 4/8 respectively. Other parameters are SF10 and SNR = -10dB.

Fig. 13d shows that SLoRa performs the best regardless of the coding rate. For example, when CR is 4/6, the decoding accuracy of SLoRa is **1.4 times** that of PLoRa, reaching 86.9%, while LoRa only decodes no more than 27% of the data. The performance gains are greater when CR is 4/5 or 4/6 because LoRa uses hamming decoding, which can only detect errors but cannot correct them under CR = 4/5 or 4/6. When CR is 4/7 and 4/8, it can correct one-bit errors but cannot handle multi-bit errors, which is not enough to eliminate the adverse effects of CTI. SLoRa decodes according to the confidence of each bit, even if CR is equal to 4/5, it is possible to correct errors. Note that in these experiments, SLoRa does not offer 100% decoding accuracy. This is because SLoRa fully follows the standard LoRa packet structure and coding algorithm. Although standard LoRa adopts interleaving to provide resilience against burst interference, it only applies interleaving for bits within one block (block is m $\times$ SF in our case and m is the codeword length). As a result, it is possible that large errors exist within one codeword, leading to demodulation errors. If the interleaving is applied at the packet level, SLoRa will offer more resilience via its soft decoding.

C. General protection

SLoRa does not need to know any prior knowledge of interference in advance. Therefore, our system adopts a general design, which can effectively recover LoRa symbols under various interference and different frequency bands.

a) Performance with different CTIs. In order to verify the universality of SLoRa under different CTIs. We also chose Zigbee and Bluetooth interference to verify the performance of SLoRa. These two kinds of interference also work in the 2.4GHz band and have certain negative effects on LoRa communication. Fig. 17a, 17b show the decoding accuracy of SLoRa under these two interference. SLoRa maintains good performance in different SF and interference settings through our framework. Note that ZigBee packets are more destructive than Bluetooth because ZigBee has a longer packet duration. In practical IOT applications, there are many scenarios where multiple technologies coexist and generate different CTI. We expect SLoRa will be able to offer resilience in general but its performance might degrade with increasing CTI levels.

b) Performance in different frequency bands. To test the general feasibility of SLoRa in different LoRa working frequency bands, we also use the LoRa SX1276 chip working at 915MHz and 433MHz. The comparison results are in Fig. 17c and Fig. 17d. Similar to the 2.4GHz, LoRa in other open spectrums also suffers from CTI. Through the symbol recovery, SLoRa offers better interference reliability.

IX. Related Work

LoRa offers long-range wireless connections with low energy consumption [20]–[24]. Operating in the unlicensed ISM band, LoRa inevitably suffers from either homogeneous interference among LPWAN devices or heterogeneous interference across different protocols [25], [26]. Wireless interference easily corrupts LPWAN packets, leading to a prolonged transmission delay and extra energy consumption.

Combating homogeneous interference: Recent studies aim at resilient LoRa communication under homogeneous interference. They can be summarized into the following two categories. The first category of literature [27]–[35] process received PHY signal to separate concurrent LoRa transmissions. These techniques commonly focus on identifying signal features (central frequency offset, frequency pattern, or phase diversity) that are unique to different LoRa devices. The second line of approaches [14] [36] [37] [38] design encoding and decoding schemes to boost the network scalability.

Combating heterogeneous interference (CTI): Different from homogeneous interference, CTI is resulted by other wireless technologies. Given the unlicensed spectrum and the LoRa's low-power nature, CTI significantly affects LoRa communication [9]–[11]. To address this problem, signal analysis techniques, e.g., PSR [12], propose partial symbol recovery via short-time Fourier transform. In addition, multiple forward error correction techniques [13], [39] add additional coding bits to correct errors in received LoRa packets.

SLoRa differs from existing literature on protection against CTI. It first proposes a lightweight symbol recovery scheme and then analyzes the confidence of recovered symbols to exploit them via soft decoding for enhanced reliability. In contrast, previous signal analysis techniques blindly treat all recovered symbols as the same regardless of specific CTI situations, in addition to their intensive computation requirement. As for existing coding solutions, they waste critical PHY-layer information, which could be exploited for a significant reliability boost. Different from the literature, SLoRa's systematic framework enables synergic interference resilience, which will be increasingly important given explosive IoT applications. It is worth noting that SLoRa's symbol recovery and confidence estimation directly rely on LoRa's unique signal characteristics under CTI and they are thus fundamentally different from previous soft decoding methods [38], [40], [41].

X. Conclusion

This paper presents SLoRa, a lightweight framework for LoRa network that achieves synergic interference resilience by integrating the symbol recovery and soft decoding. Extensive theoretical analysis and experiments based on testbeds validate the protection against CTI under various settings.

XI. Acknowledgement

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